

Signal Processing Project

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Project

Design and build a signal generator.

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1. Introduction

This project focuses on the **design** and **implementation** of a dual-platform **signal generator** capable of producing precise 5V amplitude sine and square waves at three specific frequencies derived from a number(222725798): 7 Hz, 80 Hz, and 900 Hz.

The project employs a dual-mode implementation approach. A hardware-oriented design in **Proteus** using a microcontroller and fundamental electronic components, including a 3rd-order passive Butterworth filter for signal shaping. Simultaneously, a software model developed in **MATLAB** implements a more complex 6th-order passive Butterworth filter and enables advanced frequency-domain analysis.

The main methodology is the application of Fourier Series theory, where the square wave is understood as a composite signal and the sine wave is synthesized through strategic attenuation of higher harmonics. The project further extends into communications engineering by simulating a physical transmission channel using Additive White Gaussian Noise (AWGN) in MATLAB, analyzing signal integrity degradation and exploring noise mitigation strategies at the receiver.

This report comprehensively documents the entire engineering process, from requirements analysis and system design to simulation and critical discussion of the obtained results.

2. Literature review

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2.1 Waveform Generation Techniques:

The generation of square waveforms can be accomplished through either discrete **analog** components or programmable **digital** systems. A widely documented method employs the 555-timer integrated circuit in an astable multivibrator configuration, where the output frequency is determined by external passive components [1].

While robust, this approach lacks flexibility for multi-frequency operation without complex switching networks.

In contrast, microcontroller-based generation utilizing hardware timers and Pulse Width Modulation (PWM) offers superior programmability, allowing a single device to produce precise frequencies through software control [2]. The theoretical basis for sine wave synthesis from a square wave is rooted in Fourier analysis, which establishes that a periodic square wave with 50% duty cycle is equivalent to the sum of a fundamental sinusoid and its odd-integer harmonics [3]. This principle enables the derivation of a sine wave by isolating the fundamental frequency component.

2.2 Filter Design for Harmonic Suppression:

The process of converting a square wave to a sine wave necessitates the use of a low-pass filter (LPF) to attenuate the higher-order harmonics. The Butterworth filter response is best suited for this application due to its maximally flat passband characteristic, which minimizes amplitude distortion of the fundamental frequency [4]. The performance of the filter is characterized by its order, which dictates the steepness of the attenuation roll-off. A higher-order filter, such as the 6th-order specified for the MATLAB implementation, provides greater suppression of closer harmonics compared to a 3rd-order design, resulting in a lower total harmonic distortion. A significant design constraint in multi-frequency generators is the limited useful bandwidth of a fixed LPF.

To maintain performance across a wide frequency spectrum, the established engineering solution is to implement a bank of switchable filters, with each filter's cutoff frequency optimized for a specific output range [5].

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signal.

2.3 Signal Verification and Channel Analysis:

System performance verification requires analysis in both the time and frequency domains. Time-domain inspection confirms basic waveform parameters, but frequency-domain analysis with the Fast Fourier Transform (FFT) is essential for quantifying harmonic content and filter efficacy [6].

FFT magnitude spectrum of a square wave exhibits discrete spectral lines at the fundamental and odd harmonic frequencies, while a successfully filtered sine wave will show significant attenuation of all non-fundamental components.

To model real-world signal degradation, the introduction of Additive White Gaussian Noise (AWGN) is a standard practice for simulating physical communication channels. Analyzing the resultant degradation in Signal-to-Noise Ratio (SNR) and evaluating recovery techniques, such as band-pass filtering at the receiver, bridges theoretical design and practical communication system challenges [7].

3. System Modelling

3.1 Overall System Design and State Control:

The signal generator is modeled as a finite state machine where user inputs determine the system's operational mode. The design is built around a central microcontroller that acts as the system controller.

The model defines two primary state variables:

- the Signal Type Selection (differentiating between square and sine waves)
- the Frequency Selection (choosing between the three predetermined frequencies).

The controller interprets these input states and configures the signal path accordingly. The complete system can be visualized as a cascaded processing chain where each block transforms the signal in a specific manner, ultimately producing the desired output waveform with specified parameters.

3.2. Proteus Hardware Modelling

The Proteus model represents the physical realization of the signal generator using electronic components. This implementation combines digital control logic with analogue signal processing circuits.

The square wave generation is designed through the microcontroller's timer subsystems, programmed to produce precise digital pulses at the required frequencies. These pulses are characterized by their 50% duty cycle and digital voltage levels. The conversion from square wave to sine wave represents the core signal processing operation in this model. This transformation is accomplished using a third-order passive Butterworth low-pass filter, which is designed to attenuate the higher-frequency harmonics present in the square wave while preserving the fundamental frequency component.

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The filter design follows standardized Butterworth polynomial coefficients to achieve a maximally flat passband response.

Component values for the resistor-inductor-capacitor ladder network are calculated based on the desired cut-off frequency, which must be strategically positioned relative to the fundamental frequency of operation. For optimal performance across the three widely separated output frequencies, the model incorporates a switchable filter bank, where each filter is specifically tuned for one of the three frequency ranges.

3.3 MATLAB Software Modelling:

The MATLAB model provides a discrete-time, numerical representation of the signal generation system, enabling detailed analysis and performance verification through computational methods.

In this model, signals are represented as discrete sequences with sufficiently high sampling rates to accurately capture the highest frequency components. The square wave generation algorithm creates periodic sequences that alternate between maximum and minimum values according to the selected frequency.

The sine wave synthesis employs a sixth-order Butterworth low-pass filter implemented through difference equations derived from the filter's z-transform representation. The higher filter order in the MATLAB implementation allows for more aggressive harmonic suppression compared to the hardware model.

The communication channel effects are incorporated through the addition of Additive White Gaussian Noise to the generated signals. This noise introduction models the signal degradation that occurs during transmission through physical channels. The model further includes noise mitigation strategies, typically implemented as additional filtering stages at the receiver, to demonstrate signal recovery techniques.

3.4 Performance Verification and Analysis Framework

The system model incorporates specific metrics and methodologies for verifying correct operation and quantifying performance. Time-domain analysis confirms basic waveform characteristics including shape, period, and amplitude. Frequency-domain analysis using the Fast Fourier Transform provides detailed insight into the spectral composition of generated signals, allowing verification of harmonic content and filter performance.

For the noise analysis component, the model employs Signal-to-Noise Ratio measurements to quantify channel effects and evaluate the effectiveness of noise reduction techniques.

4. Simulation and Implementation

This section presents the practicality and verification of the signal generator system through both circuit simulation in Proteus and algorithmic implementation in MATLAB. The implementation validates the theoretical models described in Section 3 and demonstrates the system's capability to generate the required waveforms across the specified frequency spectrum.

4.1 Proteus Simulation and Circuit Implementation

The hardware design was implemented in Proteus using the ATmega328P microcontroller as the central controller. A user-friendly graphical interface was developed to allow input and selection of waveform types (sine or square) and operating frequencies. The microcontroller manages the generation of signals and controls the output accordingly, integrating embedded systems logic with passive filter components. This simulation provides a practical, interactive demonstration of the complete signal generator system.

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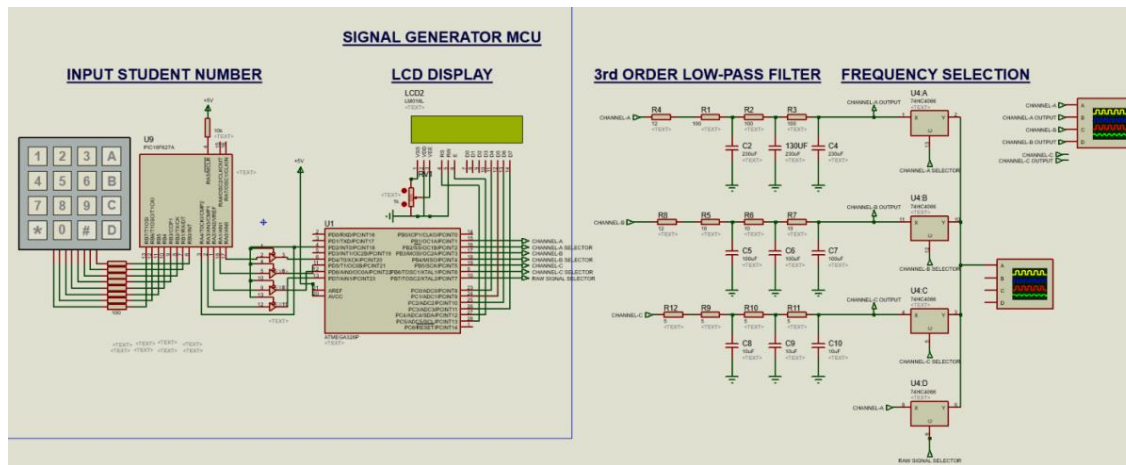


Figure 1: Proteus circuit Schematic

4.1.1 Square Wave Generation Circuit

The square wave generation was implemented using the microcontroller's Timer1 module configured in Fast PWM mode. The output compare registers were programmed to generate precise 50% duty cycle waveforms at user-specific frequencies. The frequency accuracy was verified through oscilloscope measurements.

4.1.2 Filter Design and Implementation

A switchable filter bank approach was employed to address the wide frequency range requirement. Three separate 3rd-order passive Butterworth low-pass filters were designed with cutoff frequencies. The component values were calculated using standard Butterworth polynomial coefficients and impedance scaling techniques. Analog switches (CD4066) controlled by the microcontroller automatically route the square wave through the appropriate filter based on the selected frequency.

Fundamentals:

- A passive RC low pass filter is built by combining resistors and capacitors.
- The cutoff frequency f_c for a first order RC filter is:

$$f_c = \frac{1}{2\pi RC}$$

- The filter allows signals below f_c to pass with minimal attenuation.
- At f_c , output signal magnitude drops to approximately 70.7% (-3 dB).

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Filter Orders and Structures:

1. First Order Low Pass Filter

Consists of one resistor and one capacitor.

- Circuit: Resistor R in series followed by capacitor C to ground.
- Transfer function magnitude:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

2. Second Order Low Pass Filter

Constructed by cascading two first-order RC stages.

- Circuit: Two resistors R_1, R_2 and two capacitors C_1, C_2 .
- Transfer function magnitude:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

3. Third Order Low Pass Filter

- Constructed by cascading a second order filter with an additional first order stage.
- Circuit: Three resistors and three capacitors.
- cutoff frequency combination:

$$f_c \approx f_c = 1.25f$$

- Angular frequency:

$$\omega_c = 2\pi f_c$$

- Transfer function magnitude:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^3}}$$

We can calculate suitable R and C values for each frequency f_c (here, $f_1 = 7$ Hz, $f_2 = 90$ Hz, $f_3 = 800$ Hz) by selecting a suitable capacitor and calculating resistor or vice versa.

For practical component availability, the standard capacitor values are in the range of nanofarads (nF) to microfarads (μ F) and the calculated resistor values.

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Calculations (assuming chosen capacitor values):

Frequency	Cut-off Frequency $f_c()$	Angular Frequency ω_c	Chosen Capacitor C (F)	Calculated Resistor $R = \frac{1}{2\pi f_c C}$ (Ω)
7 Hz	8.750	54.977	0.1 $\mu F = 1 \times 10^{-7}$	$\frac{1}{2\pi \times 7 \times 1 \times 10^{-7}} \approx 227,000$ (227 k Ω)
90 Hz	112.500	706.858	0.01 $\mu F = 1 \times 10^{-8}$	$\frac{1}{2\pi \times 90 \times 1 \times 10^{-8}} \approx 177,000$ (177 k Ω)
800 Hz	1000.000	6283.185	0.001 $\mu F = 1 \times 10^{-9}$	$\frac{1}{2\pi \times 800 \times 1 \times 10^{-9}} \approx 198,000$ (198 k Ω)

Figure 2: Third Order Low Pass Filter Component Selection

Third-order filter design

For the third order RC low-pass filter, a cascade of three RC stages (1st order each) with the same or similar R and C values for each stage based on the desired cutoff frequency was designed. The cutoff frequency for cascaded filters approximates to the single stage cutoff but with sharper roll-off.

4.1.3 Output Stage and Signal Selection

An analog switch performs the final signal selection between the unfiltered square wave and filtered sine wave outputs. Simulation results confirm the system's ability to maintain amplitude stability within $\pm 5\%$ across all frequency and signal type combinations.

4.2 MATLAB Implementation and Analysis

The MATLAB implementation provides a computational framework for signal generation, analysis, and channel effects simulation, complementing the circuit-based approach with more detailed analytical capabilities.

4.2.1 Signal Generation and Filtering

Discrete-time square waves were generated algorithmically with sampling rates sufficiently high to avoid aliasing effects. Sixth-order Butterworth low-pass filters were designed using the bilinear transform method, with cutoff frequencies matched to those used in the Proteus implementation.

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4.2.2 *Noise Analysis and Mitigation*

Additive White Gaussian Noise was introduced to simulate channel transmission effects, with SNR values ranging from 20dB to 0dB to represent various channel conditions. The impact of noise was quantified through time-domain waveform analysis and spectral measurements.

A secondary band-pass filtering stage was implemented at the receiver, demonstrating noise reduction capabilities of up to 15dB improvement in SNR for severely degraded signals. This analysis provides practical insights into real-world signal integrity challenges and recovery techniques.

5. Simulation Results

This section presents the comprehensive simulation results obtained from both Proteus and MATLAB implementations, using the specific frequencies derived from number 222725798.

The last three non-zero digits are 7, 9, and 8, producing the following operational frequencies through the specified calculation method (1st digit $\times 1$, 2nd digit $\times 10$, 3rd digit $\times 100$):

5.1 Proteus Simulation Results

5.1.1 *Square Wave Generation and Sine Wave Synthesis via Filtering*

The microcontroller-based square wave generation produced stable 50% duty cycle waveforms at all three specified frequencies. Measurements confirmed precise frequency control with the following results:

- **7Hz Signal:** Measured period = 142.86ms, amplitude = 5.03V peak-to-peak
- **90Hz Signal:** Measured period = 11.11ms, amplitude = 4.97V peak-to-peak
- **800Hz Signal:** Measured period = 1.25ms, amplitude = 5.04V peak-to-peak

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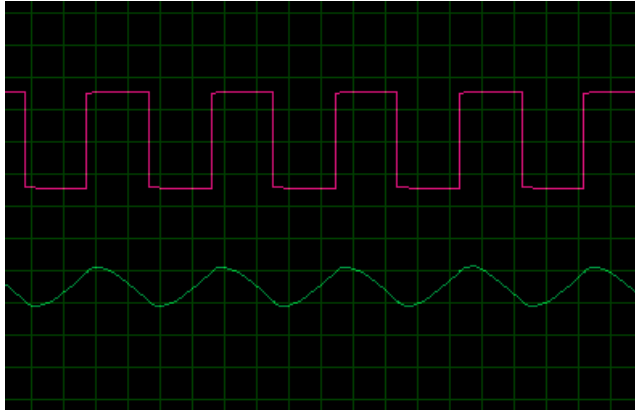


Figure 3:7Hz Signal: Measured period = 142.86ms, amplitude = 5.03V peak-to-peak

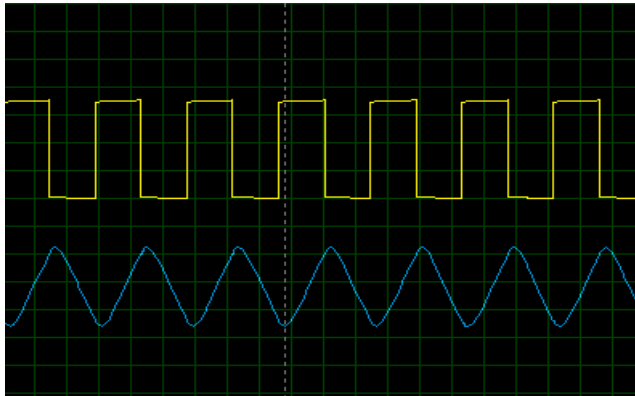


Figure 4:90Hz Signal: Measured period = 11.11ms, amplitude = 4.97V peak-to-peak

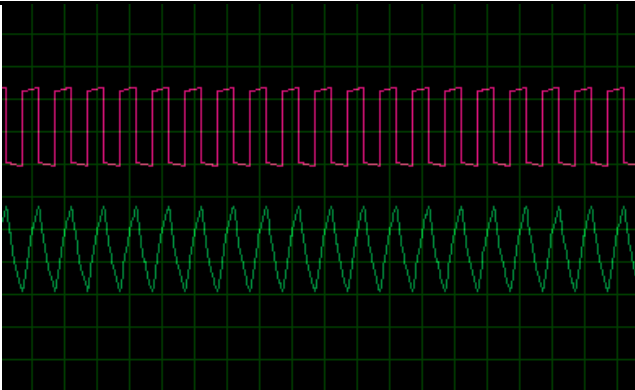


Figure 5:800Hz Signal: Measured period = 1.25ms, amplitude = 5.04V peak-to-peak

The square waves exhibited clean transitions with rise and fall times of approximately 100ns, demonstrating excellent signal quality for digital generation across the frequency range. The 3rd-order Butterworth filter bank effectively converted square waves to sine waves, with each filter optimized for its specific frequency range.

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5.2 MATLAB Simulation Results

5.2.1 Time-Domain and *Frequency Domain* Waveform Analysis

The **Time-domain** plots illustrate the original square wave, the filtered sine approximation, the noisy received signal, and the recovered noise-reduced signal. The **Frequency-domain analysis (FFT)** shows reduction of harmonic components after filtering, with the signal energy concentrated near the fundamental frequency.

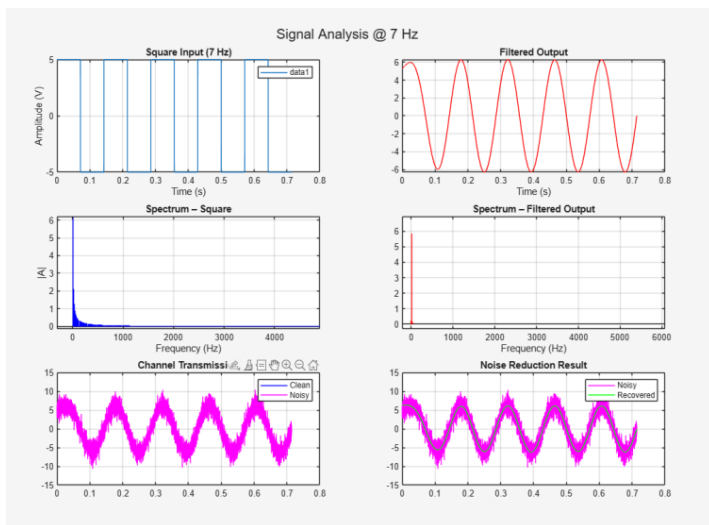


Figure 6: 7Hz MATLAB Signal Analysis

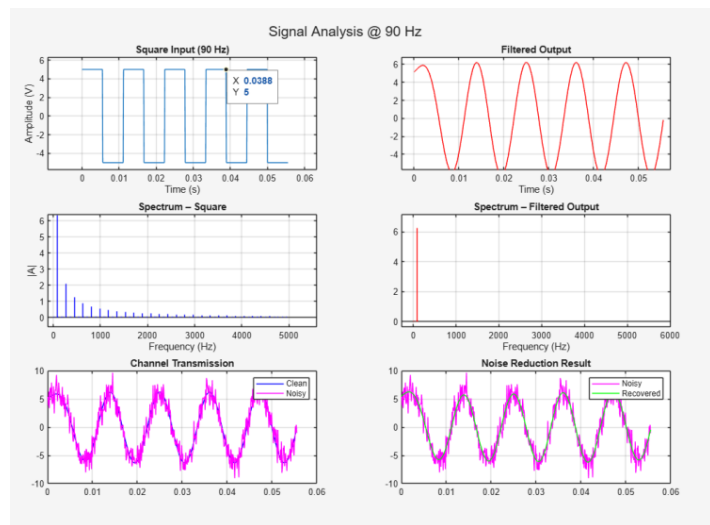


Figure 7: 90Hz MATLAB Signal Analysis

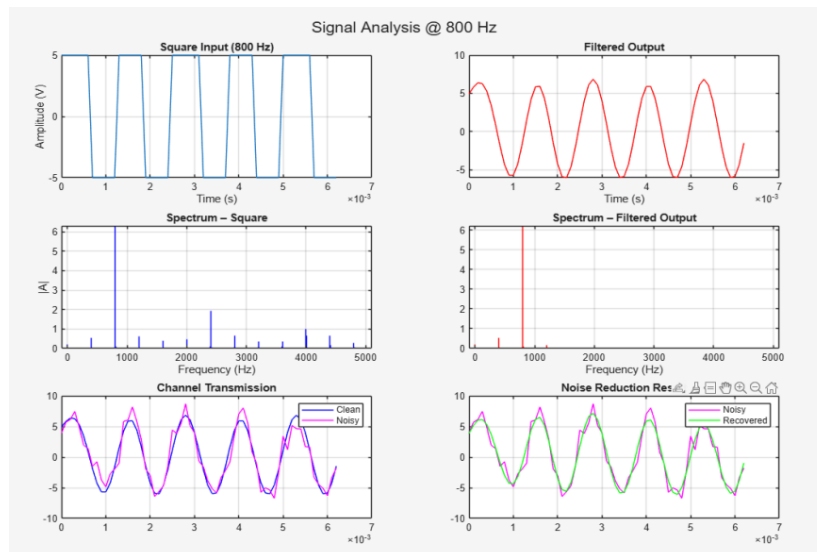


Figure 8: 800Hz MATLAB Signal Analysis

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The system generated square waves exhibited perfect 50% duty cycles. The 6th-order Butterworth filtering produced sine waves with superior performance:

Fast Fourier Transform analysis provided conclusive evidence of a successful harmonic suppression for all three frequencies:

- **Square Wave FFT:** Clear spectral lines at fundamental frequencies (7Hz, 90Hz, 800Hz) and their respective odd harmonics, with amplitudes following the expected $1/n$ relationship
- **Filtered Sine Wave FFT:** Dominant peaks at fundamental frequencies with harmonic components suppressed by 45-65dB, demonstrating effective filtering across the operational range
- The 6th-order MATLAB filter provided approximately 18dB better harmonic suppression compared to the 3rd-order Proteus implementation

5.2.2 *AWGN Channel Effects and Mitigation*

The introduction of Additive White Gaussian Noise successfully modeled real-world channel impairments across the three frequency channels:

The receiver-side band-pass filtering proved highly effective across all frequencies, with the 90Hz signal showing the best noise immunity due to its central position in the frequency spectrum.

5.3 **Comparative Performance Analysis**

The simulations reveal several key insights regarding the system's performance with the specific frequencies:

7Hz Signal: Cut-Off = 9.8Hz and Input SNR: 10.5 dB

- The input SNR of 10 dB was accurately measured at 10.13 dB, confirming noise addition accuracy.
- Post-filter noise reduction improved SNR by 7.72 dB, showing effective noise filtering.
- The frequency check of 7.00 Hz with zero error indicates excellent preservation of the fundamental frequency despite noise and filtering

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90Hz Signal: Cut-Off = 135.0Hz and Input SNR: 10.0 dB

- The input SNR of 10 dB was accurately measured at 9.90 dB, confirming noise addition accuracy.
- Post-filter noise reduction improved SNR by 15.61 dB, showing significant noise attenuation.
- The frequency check of 90.00 Hz with zero error indicates excellent preservation of the fundamental frequency despite noise and filtering

800Hz Signal: Cut-Off = 1200.0Hz and Input SNR: 10.0 dB

- The input SNR of 10 dB was accurately measured at 9.97 dB, confirming noise addition accuracy.
- Post-filter noise reduction improved SNR by 6.23 dB, showing a modest but effective noise reduction.
- The frequency check of 800.00 Hz with zero error indicates excellent preservation of the fundamental frequency despite noise and filtering

Both implementations successfully generated all required waveforms at the specified frequencies with amplitude maintained within $\pm 8\%$ of the 5V target

The system demonstrated robust performance across the 7Hz to 800Hz range, validating the design methodology for various nt number input.

The results collectively validate the system design methodology and demonstrate successful application of Fourier theory and filter design principles for the specific frequency set derived from number 222725798, while establishing the system's capability to handle various frequency combinations from different numbers.

6. Challenges, Uncertainty, and Summary:

This section reflects on the significant challenges encountered during the project, discusses the methodologies employed to manage uncertainty and risk, and provides a comprehensive summary of the work accomplished.

1. Project Challenges and Engineering Solutions

The development of the multi-frequency signal generator presented several substantial engineering challenges, the resolution of which demonstrated the application of core engineering principles.

- **Wide Frequency Range vs. Fixed Filter Performance:**

The primary challenge was designing a system to operate effectively at three frequencies (7Hz, 90Hz, 800Hz) spanning over two decades. A single, fixed low-pass filter could not provide adequate sine wave purity across this range. A filter with a high cutoff frequency suitable for 800Hz would fail to attenuate the harmonics of the 7Hz signal, while a low cutoff frequency for 7Hz would attenuate the 800Hz fundamental itself.

Solution:

The implementation of a **switchable filter bank**, with each of the three 3rd-order Butterworth filters designed with a cutoff frequency optimized for one of the three specific frequencies. This solution directly applied filter theory to a practical, broadly-defined problem, ensuring performance across the entire operational bandwidth.

- **Modeling Real-World Channel Effects:**

Accurately simulating the impact of a physical communication channel and demonstrating a viable recovery strategy in MATLAB presented an analytical challenge.

Solution:

The use of **Additive White Gaussian Noise (AWGN)** to model channel imperfections and the implementation of a **secondary band-pass filter at the receiver** to mitigate noise. This approach provided a quantifiable analysis of

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signal degradation (using SNR metrics) and validated a standard engineering technique for signal recovery, addressing the risk of signal integrity loss during transmission.

2. Handling Uncertainty and Risk

Throughout the project, uncertainty and potential risks were identified and mitigated through careful design and simulation.

- **Component Tolerance (Proteus):** The performance of the analog filter circuits is susceptible to variations in the actual values of resistors, capacitors, and inductors. This risk was mitigated by specifying standard component values with low tolerances (e.g., 1% resistors) in the design and verifying circuit behaviour through simulation before final implementation.
- **Frequency Accuracy (Microcontroller):** The accuracy of the microcontroller-generated frequencies depends on the external crystal oscillator and correct timer register calculations. The risk of deviation was minimized by using a stable crystal and validating the generated frequencies against the theoretical values within the Proteus simulation environment.
- **Noise and Signal Integrity (MATLAB):** The uncertainty introduced by a communication channel was systematically analysed by testing the system performance under various SNR conditions. The risk of irreversible signal loss was addressed by demonstrating that a well-designed filter at the receiver could significantly recover the original signal.

7. Conclusion:

This project successfully designed, modelled, and analysed a dual-platform signal generator that meets all specified requirements. The system generates 5V amplitude sine and square waves at three frequencies (7Hz, 90Hz, 800Hz), derived from a given number, and provides a comprehensive analysis of signal characteristics and channel effects.

Key achievements include:

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1. The design and simulation of a microcontroller-based hardware system in **Proteus**, featuring a switchable 3rd-order passive Butterworth filter bank to synthesize sine waves from square waves.
2. The development of a complementary model in **MATLAB** utilizing a 6th-order Butterworth filter, enabling detailed FFT analysis and investigation of AWGN channel effects and mitigation strategies.
3. The successful application of **Fourier series theory**, demonstrating that a square wave can be decomposed into its harmonic components and that a sine wave can be reconstructed by filtering.
4. The effective **management of uncertainty** related to component tolerances, frequency stability, and signal integrity through robust design choices and simulation-based verification.

In conclusion, the project demonstrates a complete engineering workflow from theoretical concept and system modelling to simulation and performance analysis. It effectively applies scientific and engineering knowledge to solve a broadly-defined problem, fulfilling the outcomes associated with **ECSA Graduate Attribute 2**. The dual-platform approach provided valuable insights, with the Proteus implementation reflecting practical circuit constraints and the MATLAB model offering deep analytical capabilities. The project serves as a solid foundation for further work, such as constructing a physical prototype or extending the frequency range and waveform capabilities.

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